

$$a) \frac{2x+3}{x^3+x^2-2x} \Rightarrow \frac{2x+3}{x(x+2)(x-1)} \Rightarrow \frac{A}{x} + \frac{B}{x+2} + \frac{C}{x-1}$$

$$2x+3 = A(x+2)(x-1) + B(x-1)x + C(x+2)x$$

$$\bullet \text{ si } x = -2 \Rightarrow -1 = 6B$$

$$B = -1/6$$

$$\bullet \text{ si } x = 1 \Rightarrow 5 = 3C$$

$$5/3 = C$$

$$\bullet \text{ si } x = 0 \Rightarrow 3 = -2A$$

$$-3/2 = A$$

$$\frac{2x+3}{x^3+x^2-2x} = -\frac{3}{2x} - \frac{1}{6(x+2)} + \frac{5}{3(x-1)}$$

$$b) \frac{x^2-8x+15}{x^2-6x+5} \Rightarrow \frac{A}{x-5} + \frac{B}{x-1}$$

$$x^2-8x+15 = A(x-1) + B(x-5)$$

$$\bullet \text{ si } x = 1 \Rightarrow 8 = 4B$$

$$2 = B$$

$$\bullet \text{ si } x = 5 \Rightarrow 25-40+15 = A4$$

$$A = 0$$

$$\frac{x^3}{5x^4 - 5} \Rightarrow \frac{A}{5(x-1)} + \frac{B}{x+1} + \frac{Cx+D}{x^2+1}$$



$$x^3 = A(x+1)(x^2+1) + 5B(x-1)(x^2+1) + (Cx+D)(5x-5)(x+1)$$

$$\text{Si } x = -1 \Rightarrow -1 = -20B$$

$$1/20 = B$$

$$x^3 = Ax^3 + Ax^2 + Ax + A + 5Bx^3 + 5Bx - 5Bx^2 - 5B + 5Cx^3 - 5Cx + 5Dx^2 - 5D$$

$$x^3 = (A + 5B - 5C)x^3 + (A - 5B + 5D)x^2 + (A + 5B - 5C)x + (A - 5B - 5D)$$

$$A + 5B - 5C = 1 \Rightarrow A + \frac{1}{4} - 5C = 1 \Rightarrow A - 5C = \frac{3}{4} \Rightarrow -5C = \frac{3}{4} - A \Rightarrow C = \frac{-\frac{3}{4} + A}{5}$$

$$A - 5B + 5D = 0$$

$$A + 5B - 5C = 0$$

$$A - 5B - 5D = 0$$

$$d) \frac{4}{x^3+x} \Rightarrow \frac{4}{x(x^2+1)} \Rightarrow \frac{A}{x} + \frac{Bx+C}{x^2+1}$$

$$4 = A(x^2+1) + (Bx+C)x$$

$$\text{Si } x=0, A=4$$

$$4 = Ax^2 + A + Bx^2 + Cx$$

$$4 = (A+B)x^2 + Cx + A$$

$$A+B=0 \Rightarrow B=-4$$

$$C=0$$

$$A=4$$

$$\left. \begin{array}{l} A+B=0 \Rightarrow B=-4 \\ C=0 \\ A=4 \end{array} \right\} \frac{4}{x^3+x} = \frac{4}{x} + \frac{-4x}{x^2+1}$$

f)

$$\frac{3x^3-8x^2+10}{x^4-3x^3+3x^2+x} = \frac{3x^3-8x^2+10}{x(x^3-3x^2+3x+1)} \Rightarrow \frac{A}{x} + \frac{Bx+C}{x^3-3x^2+3x+1}$$

$$3x^3-8x^2+10 = A(x^3-3x^2+3x+1) + (Bx+C)x$$

$$\bullet \text{ Si } x=0, 10=A$$

$$3x^3-8x^2+10 = Ax^3-3Ax^2+3Ax+A+Bx^2+Cx$$

$$3x^3-8x^2+10 = Ax^3 + (-3A+B)x^2 + (3A+C)x + A$$

$$a) \frac{x^5 - x^4 - 3x + 5}{x^4 - 2x^3 + 2x^2 - 2x + 1}$$

$$x^5 - x^4 - 3x + 5 : x^4 - 2x^3 + 2x^2 - 2x + 1 = x + 1$$

$$-(x^5 - 2x^4 + 2x^3 - 2x^2 + x) \\ x^4 - 2x^3 + 2x^2 - 4x + 5 \\ -(x^4 - 2x^3 + 2x^2 - 2x + 1) \\ -2x + 4$$

$$x + 1 + \frac{-2x + 4}{x^4 - 2x^3 + 2x^2 - 2x + 1}$$

$$\begin{array}{ccccc|c} 1 & -2 & 2 & -2 & 1 & \\ \hline & 1 & -1 & 1 & -1 & c=1 \\ \hline 1 & -1 & 1 & -1 & 0 & \\ & 1 & 0 & 1 & & c=1 \\ & 1 & 0 & 1 & 0 & \end{array}$$

$$(x^2 + 1)(x - 1)^2$$

$$\frac{-2x + 4}{(x^2 + 1)(x - 1)^2} = \frac{A}{x - 1} + \frac{B}{(x - 1)^2} + \frac{Cx + D}{x^2 + 1}$$

$$-2x + 4 = A(x - 1)(x^2 + 1) + B(x - 1) + (Cx + D)(x - 1)^2$$

$$\bullet \text{ So, } x = 1; \quad 2 = 2B, \quad B = 1$$

$$-2x + 4 = Ax^3 + Ax^2 - Ax - A + Bx^2 + B + Cx^3 - 2Cx^2 + Cx + Dx^2 - 2Dx + D$$

$$-2x + 4 = (A + C)x^3 + (-A + B - 2C + D)x^2 + (A + C - 2D)x + (-A + B + D)$$

$$A + C = 0 \Rightarrow A = -C$$

$$-A + B - 2C + D = 0 \Rightarrow 1 - C + D = 0 \Rightarrow D = C - 1$$

$$A + C - 2D = -2 \Rightarrow D = 1$$

$$-A + B + D = 4$$

$$D = 1, \quad C = 2, \quad A = -2$$

$$x + 1 - \frac{2}{x - 1} + \frac{1}{(x - 1)^2} + \frac{2x + 1}{x^2 + 1}$$

$$\frac{-5x^2 + 7x - 18}{x^4 + 6x^2 + 9}$$

$$\frac{-5x^2 + 7x - 18}{(x^2 + 3)^2} \Rightarrow \frac{Ax + B}{x^2 + 3} + \frac{Cx + D}{(x^2 + 3)^2}$$

$$-5x^2 + 7x - 18 = (Ax + B)(x^2 + 3) + Cx + D$$

$$-5x^2 + 7x - 18 = Ax^3 + 3Ax + Bx^2 + 3B + Cx + D$$

$$-5x^2 + 7x - 18 = Ax^3 + Bx^2 + (3A + C)x + (3B + D)$$

$$A = 0$$

$$B = -5$$

$$3A + C = 7 \Rightarrow C = 7$$

$$3B + D = -18 \Rightarrow D = -3$$

$$\frac{-5x^2 + 7x - 18}{x^4 + 6x^2 + 9} = \frac{-5}{x^2 + 3} + \frac{7x - 3}{(x^2 + 3)^2}$$

$$\frac{6x^2 - 15x + 16}{x^3 - 3x^2 + 4x}$$

$$\frac{6x^2 - 15x + 16}{x(x^2 - 3x + 4)} = \frac{A}{x} + \frac{Bx + C}{x^2 - 3x + 4}$$

$$6x^2 - 15x + 16 = A(x^2 - 3x + 4) + Bx^2 + Cx$$

$$\circ \text{ si } x = 0, \quad 16 = 4A$$

$$A = 4$$

$$6x^2 - 15x + 16 = Ax^2 - 3Ax + 4A + Bx^2 + Cx$$

$$6x^2 - 15x + 16 = (A + B)x^2 + (-3A + C)x + 4A$$

$$A + B = 6 \Rightarrow B = 2$$

$$-3A + C = -15 \Rightarrow C = -3$$

$$4A = 16$$

$$\frac{6x^2 - 15x + 16}{x^3 - 3x^2 + 4x} = \frac{4}{x} + \frac{2x - 3}{x^2 - 3x + 4}$$

$$\frac{1}{(2x^2 - 3x + 2)^2 (3x - 1)^2} = \frac{A}{3x - 1} + \frac{B}{(3x - 1)^2} + \frac{Cx + D}{2x^2 - 3x + 2} + \frac{Ex + F}{(2x^2 - 3x + 2)^2}$$

$$1 = A(3x - 1)(2x^2 - 3x + 2)^2 + B(2x^2 - 3x + 2)^2 + (Cx + D)(3x - 1)^2(2x^2 - 3x + 2) + (Ex + F)(3x - 1)^2$$

• Sei $x = 1/3$, $1 = \frac{-B}{81} \Rightarrow B = -81$

1	-12	55	-114	90	
	3	-27	84	90	$c = 3$
1	-9	28	-30	0	
	3	-18	30		$c = 3$
1	-6	10	0		

$$(x^2 - 6x + 10)(x - 3)^2$$

a)

x^3	$Kx^2 + 3x$	-10	
	-5	$-5K + 25$	$25K - 140$
1	$K - 5$	$-5K + 28$	$25K - 150$

$c = -5$

$$25K - 150 = 0$$

$$25K = 150$$

$$K = 6$$